

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE NAME: Database Management Systems
COURSE CODE: CSA0501

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : Experiments (High)
Assessment Tool Weightage: 40%

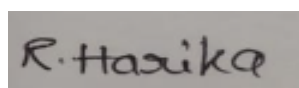
QUESTIONS		Total Marks: 50	
Q.N o	Question	Bloom's Level	Marks
1	For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.	BL2 – Understand	5
2	Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.	BL3 – Apply	5
3	Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.	BL3 – Apply	5
4	Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.	BL3 – Apply	5
5	Verify whether R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.	BL3 – Apply	5
6	Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.	BL3 – Apply	5

7	Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.	BL3 – Apply	5
8	Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).	BL3 – Apply	5
9	Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.	BL2 – Understand	5
10	Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.	BL3 – Apply	5

Code of Conduct:

I RASINENI HARIKA (192572140)(Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results

Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation
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Q1. STUDENT_COURSE Functional Dependencies and Candidate Keys

Relation:

STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade)

Assume:

- One student can take many courses.
- One course can have many students.
- SID determines student details.
- CID determines course details.
- Grade depends on both student and course.

Functional Dependencies

1. **SID $\rightarrow$ SName**
2. **CID $\rightarrow$ CName**
3. **CID $\rightarrow$ Instructor**
4. **(SID, CID) $\rightarrow$ Grade**

Therefore:

(SID, CID) $\rightarrow$ SName, CName, Instructor, Grade

Candidate Key

To determine all attributes, we need both SID and CID.

So,

Candidate Key = {SID, CID}

Conclusion

The candidate key is **(SID, CID)** because neither SID nor CID alone can uniquely identify a student-course record.

Q2. Conversion of Unnormalized Relation into 1NF

Unnormalized Relation

Suppose we have:

SID SName Courses

101 Ravi DBMS, Python

102 Arun Java, C

Here, the Courses attribute contains **multiple values**, so it is not in 1NF.

Step 1: Identify repeating groups

Courses contains more than one value for a student.

Step 2: Make every attribute atomic

Separate each course into a separate row.

1NF Relation

SID SName Course

101 Ravi DBMS

101 Ravi Python

SID SName Course

102 Arun Java

102 Arun C

Justification

A relation is in **1NF** when:

- Every attribute contains only **atomic/single values**.
- There are no repeating groups.
- Each row represents one record.

Therefore, the above relation is in **1NF**.

Q3. Decompose R(A,B,C,D,E) into 2NF

Given:

R(A,B,C,D,E)

FDs:

- $A \rightarrow BC$
- $BC \rightarrow D$
- $D \rightarrow E$

First find the key.

Step 1: Find A^+

$A^+ = \{A\}$

Using $A \rightarrow BC$:

$A^+ = \{A, B, C\}$

Using $BC \rightarrow D$:

$A^+ = \{A, B, C, D\}$

Using $D \rightarrow E$:

$A^+ = \{A, B, C, D, E\}$

Therefore:

Candidate Key = A

Step 2: Check 2NF

Since the candidate key contains only **one attribute (A)**, there cannot be a partial dependency.

Therefore, the relation is **already in 2NF**.

Decomposition

No decomposition is required for 2NF.

R(A,B,C,D,E) is in 2NF.

Note: It is not necessarily in 3NF because dependencies such as $D \rightarrow E$ can create transitive dependency.

Q4. 3NF Decomposition of EMPLOYEE

Relation:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)

Assume the following FDs:

- $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$
- $\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

Here:

$\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

This is a **transitive dependency**.

Step 1: Employee relation

Create:

EMPLOYEE(EmpID, EmpName, DeptID)

FD:

EmpID $\rightarrow$ EmpName, DeptID

Step 2: Department relation

Create:

DEPARTMENT(DeptID, DeptName, ManagerID)

FD:

DeptID $\rightarrow$ DeptName, ManagerID

Final 3NF Relations

1. **EMPLOYEE(EmpID, EmpName, DeptID)**
2. **DEPARTMENT(DeptID, DeptName, ManagerID)**

Advantage

The transitive dependency is removed, reducing:

- Update anomaly
- Insertion anomaly
- Deletion anomaly

Therefore, the schema is in **3NF**.

Q5. BCNF Verification and Decomposition

Given:

R(A,B,C,D)

FDs:

- **AB $\rightarrow$ C**
- **C $\rightarrow$ D**
- **D $\rightarrow$ A**

Step 1: Find candidate keys

(AB)⁺

AB $\rightarrow$ C

C $\rightarrow$ D

D $\rightarrow$ A

Therefore:

(AB)⁺ = {A,B,C,D}

So **AB is a candidate key**.

Also:

(BC)⁺

C $\rightarrow$ D

D $\rightarrow$ A

Therefore:

BC⁺ = {A,B,C,D}

So **BC is a candidate key**.

Similarly:

BD⁺

$D \rightarrow A$

$AB \rightarrow C$

Therefore:

$BD^+ = \{A, B, C, D\}$

So **BD is a candidate key**.

Step 2: Check BCNF

For BCNF, the left side of every FD must be a **superkey**.

- $AB \rightarrow C \rightarrow AB$ is a key 
- $C \rightarrow D \rightarrow C$ is **not a superkey**
- $D \rightarrow A \rightarrow D$ is **not a superkey**

Therefore, **R is not in BCNF**.

Decomposition

Using $C \rightarrow D$:

R1(C,D)

R2(A,B,C)

R1 is in BCNF because C is its key.

R2 has:

$AB \rightarrow C$

AB is a key of R2, so R2 is in BCNF.

Final Answer

R1(C,D)

R2(A,B,C)

Q6. Lossless-Join Decomposition Using Chase Algorithm

Consider:

R(A,B,C)

FD:

$A \rightarrow B$

Decompose into:

- R1(A,B)
- R2(A,C)

Chase Table

Create a row for R(A,B,C).

Relation A B C

R1 a b c

R2 a b₂ c

Because the FD is:

$A \rightarrow B$

Both rows have the same value for A.

Therefore, their B values must become the same.

Relation A B C

R1 a b c

R2 a b c

Now one complete row (**a,b,c**) is obtained.

Conclusion

Since the chase produces a complete row, the decomposition is **lossless**.
Therefore, $R \rightarrow R_1, R_2$ is a **lossless-join decomposition**.

Q7. MVDs and 4NF Decomposition

Relation:

TEACHER(TID, Subject, Hobby)

Suppose a teacher can have many subjects and many hobbies independently.

Therefore:

TID $\twoheadrightarrow$ Subject

and

TID $\twoheadrightarrow$ Hobby

These are **multi-valued dependencies (MVDs)**.

Problem

Suppose:

TID Subject Hobby

T1 DBMS Cricket

T1 Python Music

If subjects and hobbies are independent, all combinations are required, causing redundancy.

4NF Decomposition

Create two relations:

R1(TID, Subject)

R2(TID, Hobby)

Result

R1:

TID Subject

T1 DBMS

T1 Python

R2:

TID Hobby

T1 Cricket

T1 Music

Conclusion

The MVDs are separated, redundancy is reduced, and the relations are in **4NF**.

Q8. Join Dependency and 5NF

Join Dependency

A **join dependency (JD)** occurs when a relation can be reconstructed by joining multiple smaller relations.

Suppose:

R(Student, Course, Teacher)

A large relation may contain repeated combinations of students, courses and teachers.

This creates **redundancy**.

Example

If a student takes multiple courses and courses have multiple teachers, many combinations may be stored repeatedly.

5NF Decomposition

Decompose the relation into:

1. **STUDENT_COURSE(Student, Course)**
2. **COURSE_TEACHER(Course, Teacher)**
3. **STUDENT_TEACHER(Student, Teacher)**

The original relation can be reconstructed by joining these relations when the join dependency holds.

Advantages

- Removes join-based redundancy.
- Reduces unnecessary repeated data.
- Prevents modification anomalies.
- Produces **5NF (Project-Join Normal Form)**.

Conclusion

5NF decomposes a relation when a join dependency causes redundancy and the relation can be safely reconstructed from smaller relations.

Q9. College Database Anomalies and 3NF

Suppose a poorly designed relation is:

COLLEGE(StudentID, StudentName, CourseID, CourseName, DeptID, DeptName)

Functional Dependencies

- $\text{StudentID} \rightarrow \text{StudentName}$
- $\text{CourseID} \rightarrow \text{CourseName}, \text{DeptID}$
- $\text{DeptID} \rightarrow \text{DeptName}$

Anomalies

1. Insertion Anomaly

We cannot insert a new department unless a student or course is also added.

2. Update Anomaly

If a department name changes, it must be changed in many records.

3. Deletion Anomaly

Deleting the last student of a course may also delete important course/department information.

3NF Decomposition

Create:

STUDENT(StudentID, StudentName)

COURSE(CourseID, CourseName, DeptID)

DEPARTMENT(DeptID, DeptName)

Result

The transitive dependency:

$\text{CourseID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$

is removed.

Conclusion

The database is normalized to **3NF**, reducing insertion, deletion and update anomalies.

Q10. Attribute Closure and Candidate Keys

Attribute Closure

The **closure of an attribute set** is the set of all attributes that can be determined from it using the given functional dependencies.

Example

Let:

R(A,B,C,D)

FDs:

- $A \rightarrow B$
- $B \rightarrow C$
- $C \rightarrow D$

Find A^+ .

Step 1

Start with:

$A^+ = \{A\}$

Using $A \rightarrow B$:

$A^+ = \{A, B\}$

Using $B \rightarrow C$:

$A^+ = \{A, B, C\}$

Using $C \rightarrow D$:

$A^+ = \{A, B, C, D\}$

Therefore:

$A^+ = \{A, B, C, D\}$

Since A^+ contains all attributes of R:

A is a candidate key.

General Method

To find candidate keys:

1. Select an attribute set.
2. Calculate its closure.
3. If closure contains all attributes, it is a **superkey**.
4. Check minimality.
5. If no attribute can be removed, it is a **candidate key**.

Conclusion

Candidate Key = A